9.22 Solution: (a) For a TE mode, or the magnetic multipole field, the electric field in the cavity is given by Eq. (9.116), with $g_l(kr) = j_l(kr)$. This solution is transverse and thus has only θ andϕ components. The metallic boundary with infinite conductivity requires that the θ andϕ components of the electric field vanishes at $r = a$, which means

$$j_l(ka) = 0.$$

Let x_{ln} be the n-th solution to $j_l(x) = 0$, then the characteristic frequencies are given by

$$\omega_{lm,n} = \frac{c}{a} x_{ln},$$

independent of m.

Similarly, for the TM mode, the magnetic and the electric fields are given by Eq. (9.118), again with $g_l(kr) = j_l(kr)$. From Eq. (10.60),

$$\nabla \times \big(f_l(kr)\mathbf{X}_{lm}\big) = \frac{i\mathbf{n}\sqrt{l(l+1)}}{r} f_l(kr) Y_{lm} + \frac{1}{r}\frac{\partial}{\partial r}\big[r f_l(r)\big]\mathbf{n} \times \mathbf{X}_{lm},$$

where

$$\mathbf{X}_{lm} = \frac{\mathbf{L}Y_{lm}}{\sqrt{l(l+1)}}.$$

We can see that the transverse components of the electric field are given by the second term. Again, we require that the transverse components vanish at the boundary, which leads to

$$\frac{\partial}{\partial r}\big[r f_l(r)\big]\Big|_{r=a} = 0.$$

Let x'_{ln} be the n-th solution to the equation

$$\frac{d}{dx}\big[x j_l(x)\big] = 0,$$

then the characteristic frequencies are

$$\omega'_{lm,n} = \frac{c}{a} x'_{ln}.$$